Classical Field Theory

Lecture Notes

Fields, symmetries, variational principles, and physical laws.

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Introduction

Purpose and Character of These Notes

Classical field theory is often presented in an extremely compressed form. A few symbols are introduced, an action is written down, a variation is performed in one line, and a field equation appears. To an experienced reader, such a presentation may be efficient. To a student meeting the subject for the first time, it can make the theory look like a collection of formulas that arrive without a clear reason, operate by hidden rules, and disappear before their meaning has been examined.

These notes are designed to avoid that style of exposition. Their purpose is not merely to collect standard formulas from analytical mechanics, special relativity, electrodynamics, and gauge theory. Their purpose is to show how the main constructions of classical field theory are built, why they are introduced, what assumptions they use, how the calculations are carried out, and how the final expressions are interpreted physically and mathematically.

A formula will therefore not be treated as the end of an explanation. Whenever possible, it will be presented as one stage in a sequence:

1. identify the physical or mathematical problem;
2. determine what objects must be introduced;
3. state the assumptions and conventions;
4. derive the relevant relation;
5. test it on a simple model;
6. interpret the result;
7. examine limiting cases and possible failures.

The central principle of the course is that a student should not only recognize an equation, but should know what to do with it. For example, it is not enough to state the Euler–Lagrange field equations. A student should see how the variation of a field is defined, how derivatives of the variation arise, where integration by parts is used, why a boundary term disappears, how indices are managed, and how the resulting equation is applied to an explicit Lagrangian density. Similarly, it is not enough to write down Noether's theorem. The theorem must be used to construct an actual current, verify its divergence, and identify the associated conserved charge.

The same standard will be applied to electrodynamics. The electromagnetic field tensor will not be introduced as an isolated matrix to memorize. It will be built from the four-potential, its components will be related explicitly to the electric and magnetic fields, its transformation properties will be examined, and its invariants will be calculated. Maxwell's equations will then be derived from an action, rather than merely restated in covariant notation.

These notes are meant to serve two related purposes. During the lecture, they provide a structured path through the material and a reliable source of notation, derivations, and examples. After the lecture, they should function as a readable reconstruction of the argument. A student who did not follow one step on the board should be able to return to the notes and see how that step fits into the complete calculation.

The exposition will therefore contain several recurring elements.

Motivation before formalism

A new object will be introduced because it solves a problem or expresses a principle. The field concept will be motivated by locality and continuous degrees of freedom. The action will be motivated as a compact way to encode dynamics. Four-dimensional notation will be motivated by Lorentz symmetry. Gauge freedom will be motivated by the non-uniqueness of the potentials that describe the same physical electromagnetic field.

Definitions connected to examples

Definitions will be followed by concrete cases. Scalar fields will be represented by temperature, density, or a real dynamical field. Vector fields will be connected with velocity fields and electric fields. Tensor fields will be introduced through objects that are actually used later, rather than through formal classification alone.

Complete derivations

Important formulas will be derived in sufficient detail that the logical structure is visible. This does not mean that every elementary algebraic step must be expanded indefinitely. It means that no essential conceptual step should be hidden behind phrases such as "it is easy to show" or "after some manipulation." When a sign, boundary condition, index contraction, or change of variables matters, it will be displayed.

Simple proofs and checks

Not every statement in physics requires a theorem-proof format, but important claims should be justified. We will prove elementary identities when they clarify the structure of the theory, verify conservation laws directly, check gauge invariance explicitly, and compare covariant formulas with familiar three-dimensional expressions. Dimensional analysis, symmetry checks, and limiting cases will be used as routine diagnostic tools.

Worked calculations

The notes will include explicit calculations rather than only general rules. A free scalar field will be solved by a plane-wave ansatz. A Noether current will be computed for a specified transformation. The electromagnetic field tensor will be written component by component. The Green function for the Poisson equation will be applied to a point source. The radiation field of an oscillating dipole will be developed far enough to calculate the angular distribution and total emitted power.

Physical interpretation

After a derivation, we will ask what the result means. What quantity is conserved? What propagates? Which variables are measurable? Which variables contain redundancy? What changes under a transformation, and what remains invariant? A correct calculation without interpretation is incomplete, because field theory is not only a formal language but also a way of organizing physical information.

Connections between chapters

The course is designed as a chain of constructions. The action principle for particles becomes the action principle for fields. Translation symmetry becomes energy-momentum conservation. The scalar field provides a simple model before the electromagnetic field is introduced. The electromagnetic potential becomes the first example of a gauge connection. The field strength becomes the first example of curvature. These connections will be stated explicitly so that the course does not appear as twelve unrelated topics.

The level of the notes is therefore deliberately intermediate between a concise reference text and a fully expanded textbook. They will remain mathematically precise, but they will also explain why the mathematics is being used. They will contain derivations and examples, but they will avoid turning every section into a long catalogue of exercises. The objective is a coherent set of lecture notes that can be read, checked, and used.

How Each Topic Will Be Developed

To keep the exposition consistent, most substantial topics will be developed according to a common pattern. The details will vary, but the reader should repeatedly encounter the same sequence of questions: what problem are we solving, what structures are needed, how is the result derived, and how can it be used?

1. The motivating problem

A section should begin with a recognizable task or conceptual difficulty. We may want to describe a continuous system, derive a differential equation from an action, express a conservation law locally, solve a field equation with a source, or understand why two different potentials represent the same electromagnetic field. This opening question determines the purpose of the formalism that follows.

The motivation should be specific enough to guide the calculation. Instead of saying only that fields are important, we will ask how a system with one degree of freedom at every point in space can be described. Instead of saying only that symmetries imply conservation laws, we will ask how an infinitesimal transformation changes the action and how that change produces a divergence equation.

2. The objects, assumptions, and conventions

Before calculating, the relevant objects must be defined. Their domains, codomains, indices, units, transformation rules, and regularity assumptions should be clear. In relativistic calculations, the metric signature and index conventions must be fixed. In variational calculations, the allowed variations and boundary conditions must be stated. In Green-function methods, the differential operator and boundary conditions must be identified.

This stage is essential because many apparent contradictions in field theory are actually differences of convention. A sign in the metric changes signs in scalar products. A definition of the field-strength tensor changes the placement of signs in its components. A choice of units changes the constants appearing in Maxwell's equations. The notes will not hide these choices.

3. The derivation

The main relation will then be derived step by step. The derivation should show every operation that carries conceptual information. If a derivative is moved from a variation to another factor, the integration by parts will be written. If dummy indices are renamed, the reason will be visible. If antisymmetry causes a term to vanish, the antisymmetry will be used explicitly. If a boundary term is discarded, the corresponding boundary condition will be stated.

A representative example is the field-theoretic action

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu \phi, x)\, \mathrm{d}^4 x.$$

The notes will not jump directly from this expression to the Euler–Lagrange field equation. They will define a varied field $\phi + \varepsilon\eta$, differentiate with respect to ε, separate the variation of the field from the variation of its derivative, integrate by parts, analyze the boundary contribution, and only then infer the differential equation.

4. A proof, verification, or structural check

Once a formula has been derived, it will be checked. The nature of the check depends on the result. A proposed current may be shown to have zero divergence on solutions of the field equation. A covariant Maxwell equation may be expanded into its three-dimensional components. A gauge transformation may be applied directly to verify that the field tensor remains unchanged. A stress-energy tensor may be tested by computing its energy-density component.

These checks serve two purposes. They confirm the algebra, and they reveal what the abstract expression contains. A tensor formula becomes much easier to understand after one calculates its individual components and recovers a familiar physical law.

5. A fully worked example

General formulas become useful only when they are applied. Each major construction should therefore be followed by at least one complete example, and usually by several examples of increasing complexity.

For a scalar field, we may begin with the free massless field, derive the wave equation, substitute a plane wave, and obtain the dispersion relation. We may then add a mass term and compare the result. For Noether's theorem, we may first treat spacetime translations and then a global phase transformation of a complex scalar field. For Green functions, we may begin with a point source and then pass to a distributed source by integration.

A worked example should include the setup, the calculation, and the interpretation. It should not end immediately after the final algebraic expression. The reader should be told what the answer says about propagation, conservation, symmetry, or physical measurement.

6. Limiting cases and alternative forms

An important result should be examined in limits where its meaning is already known. A relativistic formula may be reduced to a nonrelativistic one. A massive scalar field may be compared with the massless case. A retarded solution may be compared with a static solution. A tensor equation may be rewritten in vector notation.

Alternative forms will also be compared when this improves understanding. The electromagnetic field may be described using $\mathbf{E}$ and $\mathbf{B}$, the four-potential A_μ, the antisymmetric tensor $F_{\mu\nu}$, or a differential two-form. These descriptions are not competing theories; they emphasize different structural features of the same theory.

7. Common mistakes and diagnostic questions

Where a calculation is especially sensitive to signs, indices, or assumptions, the notes will identify typical errors. Examples include confusing raised and lowered components, varying a derivative incorrectly, discarding a boundary term without justification, treating gauge-dependent quantities as observables, or using a static Green function for a time-dependent problem.

The reader should learn not only a successful calculation, but also how to detect an unsuccessful one. Useful questions include:

- Are the free indices the same on both sides of the equation?
- Do the dimensions agree?
- Is the expression invariant under the symmetry it is supposed to respect?
- Does the result reduce correctly in a simple limit?
- Have all boundary conditions been used consistently?
- Is the conserved quantity really independent of time?

8. A concise conclusion

A section should close by identifying what has been established and what will be used next. This final paragraph is important in a cumulative subject. The student should know which result is a definition, which is a derived equation, which is a conservation law, and which is a method of solution.

The notes will therefore distinguish clearly between several kinds of statements:

- definitions that introduce notation or structure;
- assumptions that restrict the model;
- identities that hold algebraically;
- field equations that hold for physical solutions;
- conservation laws that follow from symmetries;
- examples that illustrate a method;
- approximations that hold only in a specified regime.

This distinction prevents formulas from appearing as an undifferentiated list. It allows the reader to understand not only what an equation says, but why it is valid and when it may be used.

Course Structure and Chapter Roadmap

The course is organized into twelve principal chapters, corresponding approximately to twelve teaching weeks. The chapters are not independent units. Each one introduces tools that are used later, and the order is chosen so that the formalism grows from familiar mechanics toward relativistic fields, electrodynamics, propagation, radiation, and geometric field theories.

The first three chapters establish the field concept and the variational method. Chapters 4–6 introduce the relativistic language, the scalar-field prototype, and the relation between symmetry and conservation. Chapters 7–9 develop electromagnetism as a complete classical field theory, including gauge freedom and energy-momentum. Chapters 10–11 explain how fields generated by sources are solved and how radiation emerges. Chapter 12 gives a geometric outlook toward gauge theory and gravity.

Chapter 1: Fields, Locality, and Continuous Degrees of Freedom

The opening chapter establishes what a field is and why fields are required. We begin with familiar examples: temperature as a scalar field, the velocity of a fluid as a vector field, the displacement of a string as a time-dependent field, and the electric field as a physical quantity defined throughout space. These examples make clear that a field assigns data to every point of a domain and therefore carries infinitely many degrees of freedom.

The idea of locality is then introduced. A local field theory describes changes at a point in terms of the field and its derivatives near that point. This contrasts with a direct action-at-a-distance description. The distinction will be illustrated through sources, responses, and propagation. We will ask what it means for a source to affect a field locally and how information can move through a continuous medium.

A central worked construction will be the passage from a discrete chain of coupled oscillators to a continuous string. The discrete labels will be replaced by a spatial coordinate, finite differences will become derivatives, and the sum over masses will become an integral. This example will show explicitly how a field theory can emerge as the continuum limit of a many-particle system.

By the end of the chapter, the student should be able to identify scalar, vector, and tensor fields, state their domains and values, distinguish discrete from continuous degrees of freedom, and explain why local differential equations are natural equations of motion for fields.

Chapter 2: Action Principles: From Particles to Fields

The second chapter develops the variational principle in the setting of ordinary mechanics. This is necessary because the field-theoretic action is a direct extension of the mechanical action. We introduce a functional as an object whose input is an entire trajectory rather than a single number. The action is then defined as an integral of the Lagrangian along a path.

The variation of a trajectory will be constructed explicitly. We will compare a physical trajectory with nearby trajectories that share fixed endpoints, differentiate the action with respect to the variation parameter, and use integration by parts. The boundary term will not simply be discarded; the endpoint conditions that make it vanish will be stated and interpreted.

The Euler–Lagrange equations will be derived in full and applied to several concrete systems: a free particle, a particle in a potential, the harmonic oscillator, and a simple constrained coordinate. These examples will show how to move from a Lagrangian to an equation of motion and then verify that a proposed solution satisfies the resulting equation.

This chapter also establishes a working habit that will recur throughout the course: whenever an equation is derived from an action, the allowed variations, the boundary data, and the variables held fixed must be specified.

Chapter 3: Lagrangian Field Theory and Field Equations

The third chapter performs the transition from trajectories to fields. The Lagrangian becomes a Lagrangian density, and the action becomes an integral over spacetime. A variation now changes the field value at every point, and derivatives of the field produce derivatives of the variation.

The Euler–Lagrange field equations will be derived carefully for one field and then generalized to several coupled fields. The calculation will display the spacetime integration by parts and the boundary term on the boundary of the integration region. This provides the basic method used in nearly every later derivation.

Several field equations will be obtained from actions. The vibrating string will produce the wave equation. A static energy functional will lead to Laplace or Poisson equations. A field with a potential $V(\phi)$ will provide a first example of a nonlinear field equation. We will compare the roles of kinetic, gradient, source, and potential terms in a Lagrangian density.

The chapter will close by identifying canonical field momenta and the Hamiltonian density at an introductory level. The main goal, however, is practical: given a Lagrangian density, the student should be able to vary it, manage the boundary term, and derive the corresponding differential equation.

Chapter 4: Spacetime, Lorentz Symmetry, and Covariant Notation

Classical relativistic field theory requires a language in which space and time are combined consistently. This chapter introduces Minkowski spacetime, events, the spacetime interval, proper time, and Lorentz transformations. The metric convention will be fixed and used throughout the remaining chapters.

Four-vectors, covectors, and tensors will be introduced through calculations rather than definitions alone. Students will practice raising and lowering indices, contracting tensors, distinguishing free and dummy indices, and checking whether an expression is a Lorentz scalar. Four-position, four-velocity, four-momentum, and the four-gradient will provide the main examples.

The d'Alembert operator will be constructed from the metric and the four-gradient. Familiar equations such as the wave equation will then be rewritten in covariant form. We will compare the compact tensor notation with the expanded time-and-space components so that the notation does not conceal the underlying differential equation.

Particular attention will be given to sign conventions. The notes will show how the chosen metric signature affects norms, derivatives, and Lagrangian densities. By the end of the chapter, the student should be able to read and manipulate the index notation used in the remainder of the course.

Chapter 5: The Real Scalar Field

The real scalar field is the simplest complete relativistic field theory and will serve as the principal training model. We construct the free scalar-field Lagrangian from Lorentz scalars and derive the massless and massive field equations. The massive equation is the classical Klein–Gordon equation, treated here as a classical partial differential equation rather than as a quantum equation.

Plane-wave solutions will be substituted explicitly into the field equation. This calculation will produce the dispersion relation and clarify the meaning of frequency, wave vector, and the mass parameter. The massless and massive cases will be compared, including phase and group behavior where appropriate.

The chapter will then examine energy and momentum for scalar fields. We will calculate the energy density of simple configurations and ask under what conditions the energy is bounded below. Interacting theories with a potential $V(\phi)$ will be introduced, and equilibrium configurations will be identified as extrema of the potential.

Linearization around an equilibrium will show how a nonlinear theory produces an approximate linear wave equation for small perturbations. This provides a useful method that reappears in many areas of physics: find a background solution, perturb it, keep the leading terms, and analyze stability.

Chapter 6: Symmetries, Currents, and Noether's Theorem

This chapter develops one of the central organizing principles of field theory: continuous symmetries produce local conservation laws. We begin with infinitesimal transformations of coordinates and fields, making clear what changes actively and what changes because the coordinates have been relabeled.

Noether's theorem will be derived for fields with enough detail to identify every term in the current. The derivation will distinguish variations that vanish at the boundary from transformations that represent physical symmetries. The resulting divergence equation will be related to a conserved charge by integration over space.

Several explicit applications will follow. Time translation will be connected with energy, spatial translation with momentum, and rotations with angular momentum. A complex scalar field will provide an example of an internal global phase symmetry and its associated current. For each case, the current will be calculated and its conservation checked using the field equation.

The chapter will also prepare the transition to gauge theory by distinguishing a global symmetry from a local redundancy. The student should leave with a practical algorithm: specify the infinitesimal transformation, compute the variation of the Lagrangian density, identify the Noether current, and verify its divergence.

Chapter 7: Electromagnetic Potentials and Gauge Freedom

Electromagnetism is introduced through potentials because the potentials are the natural variables of the action principle. We begin with the scalar and vector potentials and derive the electric and magnetic fields from them. The relations will be checked against the homogeneous

Maxwell equations.

Gauge transformations will then be performed explicitly. The potentials change, but the electric and magnetic fields remain unchanged. This calculation provides the first concrete example of a description containing more variables than there are physical degrees of freedom. The distinction between a change of description and a change of physical state will be emphasized.

The scalar and vector potentials will be combined into the four-potential A_μ. From it we construct the antisymmetric electromagnetic field tensor $F_{\mu\nu}$. The matrix of components will be written out, and the electric and magnetic components will be identified with careful attention to signs and index positions.

The two standard Lorentz invariants of the electromagnetic field will be calculated and interpreted. Finally, the homogeneous Maxwell equations will be written in covariant form and related to the fact that the field tensor is constructed from a potential.

Chapter 8: Maxwell Theory from an Action

This chapter builds Maxwell theory as a variational field theory. The electromagnetic Lagrangian density is formed from the field tensor, and a coupling term to an external four-current is added. We then vary the action with respect to the four-potential.

The calculation will be shown in full: the variation of the field tensor, the use of antisymmetry, the integration by parts, the treatment of the boundary term, and the extraction of the inhomogeneous Maxwell equations. The covariant result will be expanded into Gauss's law and the Ampere–Maxwell law.

Gauge invariance of the source coupling will be analyzed. We will show that the action changes by a boundary term only when the current is conserved. This makes the relation between gauge invariance and charge conservation explicit rather than merely asserted.

The Lorenz gauge condition will be introduced and used to reduce the equations for the potentials to wave equations. The action for a relativistic charged particle interacting with the electromagnetic potential will then be varied to obtain the Lorentz-force law. This completes a unified action-based description of both the field and its interaction with matter.

Chapter 9: Energy, Momentum, and Stress in Field Theory

Field equations describe evolution, but they do not by themselves show how energy and momentum are distributed and transported. This chapter develops the local energy-momentum description of scalar and electromagnetic fields.

We begin with canonical field momentum and Hamiltonian density. Translation symmetry is then used to construct the canonical energy-momentum tensor. The distinction between canonical and symmetric tensors will be discussed, including why the symmetric form is especially natural in relativistic theories and in coupling to gravity.

For the scalar field, the energy density, momentum density, and flux will be computed. For the electromagnetic field, the components of the energy-momentum tensor will be identified with electromagnetic energy density, the Poynting vector, momentum density, and the Maxwell stress tensor.

Several explicit calculations will show how these quantities are used. We will calculate energy flow in a plane wave, force from the stress tensor in a simple geometry, and radiation pressure. Local conservation equations will be integrated over a finite region to obtain balance laws with flux through the boundary.

Chapter 10: Green Functions, Sources, and Field Propagation

The tenth chapter turns from deriving field equations to solving them. Green functions provide a systematic method for linear differential equations with sources. We introduce the idea through an operator equation and define a Green function as the response to a point source.

The static case will be developed first. The Green function of the Poisson equation in three-dimensional space will be used to recover the Coulomb potential. The solution for a distributed source will then be obtained by superposition. This calculation will connect the abstract Green-function definition with a familiar physical result.

Time-dependent equations require additional choices. We will construct retarded and advanced Green functions for wave equations and explain the boundary conditions encoded by each choice. The retarded solution will be associated with causal propagation from past sources to later fields.

The method will then be applied to electromagnetic potentials. Retarded potentials will be derived as integrals over the source evaluated at retarded time. The chapter will emphasize that a differential equation alone does not determine a unique physical solution; boundary and causal conditions are part of the problem.

Chapter 11: Radiation from Time-Dependent Sources

Radiation is the regime in which a time-dependent source creates a field that carries energy to arbitrarily large distances. We begin by separating near-field terms from radiation-field terms according to their dependence on distance. This makes clear why a $1/r$ field can transport a finite amount of energy through a large sphere.

A multipole expansion will organize the radiation produced by localized sources. The electric dipole will be studied in detail. Starting from an oscillating dipole moment, we will calculate the far-zone fields, the Poynting vector, the angular distribution of emitted power, and the total radiated power.

The Lienard–Wiechert potentials will then be introduced for a moving point charge. Their dependence on retarded time will be analyzed, and the velocity and acceleration contributions to the field will be distinguished. The acceleration field will be identified as the radiative part.

The Larmor formula will be derived or motivated from the radiation field, with its assumptions stated clearly. Energy balance and radiation reaction will be discussed at an introductory level, including the conceptual difficulties that arise when a point charge interacts with its own field.

Chapter 12: Gauge Fields and Gravity: The Geometric Outlook

The final chapter places the preceding material in a broader geometric framework. Electromagnetism will be reinterpreted as a $U(1)$ gauge theory. The four-potential will be viewed as a connection, the field tensor as curvature, and the gauge-covariant derivative as the derivative appropriate for charged fields.

This language will be introduced through the electromagnetic case before any non-Abelian generalization is attempted. The non-Abelian field strength will then be presented as a natural extension in which the gauge potentials themselves interact because the symmetry group is noncommutative.

Gravity will be introduced as a theory in which the metric is a dynamical field. Connections, curvature, geodesics, and the Einstein–Hilbert action will be discussed at a structural level. The variation of the gravitational action will be outlined sufficiently to show why the Einstein tensor appears and why stress-energy acts as a source.

The weak-field form $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ will connect general relativity with linear field theory and the Newtonian potential. The chapter will conclude by comparing electromagnetism and gravity: their dynamical variables, symmetries, sources, field strengths, conservation laws, and geometric interpretations. The purpose is not to complete a course in Yang–Mills theory or general relativity, but to show where the methods developed in the previous eleven chapters lead.

How to Read and Extend These Notes

These notes are intended to complement a lecture presented at the board. The lecture can emphasize the main physical argument, develop a representative derivation, and respond to questions. The written notes can preserve the complete structure: definitions, conventions, intermediate steps, calculations, checks, diagrams, and conclusions. Neither form should merely duplicate the other. Together they should allow the student to see both the main line of reasoning and the details needed for independent study.

A productive way to read a chapter is to separate three levels of understanding.

1. **Conceptual level:** What physical or mathematical problem is being addressed? What new object or principle is introduced?
2. **Derivational level:** How does the main equation follow from the assumptions? Which steps use algebra, calculus, symmetry, or boundary conditions?
3. **Operational level:** Given a specific field, source, or transformation, can the method be applied to obtain a concrete result?

The student should not try to memorize every displayed equation at the first reading. It is more useful to identify the role of each equation. Is it a definition, an identity, an equation of motion, a conservation law, a gauge choice, or a solution under specified boundary conditions? Once that role is understood, the formula is easier to reconstruct and use.

Calculations should be read actively. Before following a derivation, the reader should note the variables, free indices, dimensions, and expected form of the answer. After the derivation, the result should be checked in a simple case. For instance, a covariant equation can be expanded into components, a conservation law can be integrated over space, and a relativistic expression can be examined in a low-velocity limit.

The notes will be developed incrementally. Each chapter is divided into section files, and the substantive content will be written one section at a time. This is not only a repository-management choice. It supports mathematical and pedagogical quality. A single section can be checked for logical continuity, notation, sign conventions, examples, and links to earlier material before the next section is added.

The intended development cycle for future work is:

1. select one section;
2. identify its precise learning objective;
3. write the motivation and definitions;
4. develop the derivation or central argument;
5. add one or more explicit examples;
6. verify notation, equations, and limiting cases;
7. compile and inspect the PDF;
8. continue to the next section only after the current one is coherent.

This section-by-section method prevents a chapter from becoming a long sequence of loosely connected formulas. It also makes it possible to revise the course structure without rewriting unrelated material. If a topic later requires more detail, its section can be expanded or divided. If a topic proves secondary, it can be shortened without disturbing the rest of the chapter.

The twelve chapters describe the intended logical route, but the final notes should remain responsive to the actual needs of students. Some ideas may require additional preliminary explanations. Some calculations may need a second example. A diagram may communicate a geometric relation more effectively than another paragraph. The structure should therefore be stable enough to guide the course, but flexible enough to improve as the material is taught and reviewed.

A successful use of these notes should lead to more than recognition. After studying a topic, the student should be able to explain the problem in words, reproduce the main derivation with stated assumptions, carry out a representative calculation, interpret the result, and recognize the most common mistakes. That standard will guide the construction of every subsequent section.

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Chapter 2

Action Principles: From Particles to Fields

Chapter 3

Lagrangian Field Theory and Field Equations

Chapter 4

Spacetime, Lorentz Symmetry, and Covariant Notation

Chapter 5

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